Experiment 5

You are required to perform a series of tasks to explore **Git as a Distributed Version Control System (DVCS)** and demonstrate its advantages over centralized version control systems. Execute all steps carefully and record your observations wherever required.

Task 1: Repository Initialization

1. Create a new directory named dvcs-lab.
2. Initialize it as a Git repository.
3. Create a file file.txt and add initial content.
4. Stage and commit the file with an appropriate message.

Deliverable:

- Commands used
- Screenshot or output of the initial commit

Task 2: Demonstrating Distributed Nature

1. Clone the repository into a new directory named dvcs-lab-clone.
2. Verify that the cloned repository contains the full commit history.
3. Make changes in the cloned repository and commit them **without connecting to any remote server**.

Deliverable:

- Commands used
- Output of git log
- Explanation of how this task demonstrates DVCS

Task 3: Branching and Merging

1. Create a new branch named feature-branch.
2. Switch to the branch and modify file.txt.
3. Commit the changes.

4. Switch back to the main branch and merge the feature branch.

Deliverable:

- Commands used
- Merge result
- Explanation of why branching is efficient in Git

Task 4: Simulating Collaboration

1. Add a remote repository (e.g., GitHub).
2. Push your local repository to the remote.
3. Pull the changes into the cloned repository.

Deliverable:

- Commands used
- Explanation of push and pull operations

Task 5: Exploring Commit History

1. Display the commit history in a compact format.
2. Analyze the sequence of commits.

Deliverable:

- Output of git log --oneline
- Brief explanation of commit history

Task 6: Undo Changes using git revert

1. Introduce an incorrect change in file.txt and commit it.
2. Use git revert to undo the change.
3. Verify the file content and commit history.

Deliverable:

- Commands used
- Output of git log before and after revert
- Explanation of how git revert works

Task 7: Undo Changes using git reset

Perform the following operations and observe their effects:

(a) Soft Reset

- Undo the last commit using --soft

(b) Mixed Reset

- Undo another commit using default reset

(c) Hard Reset

- Undo a commit using --hard

Deliverable:

- Commands used
- Observations for each reset type
- Differences in staging area and working directory

Task 8: Comparative Analysis

Based on your experiment, answer the following:

1. How does Git demonstrate characteristics of a DVCS?
2. Compare DVCS and centralized VCS in terms of:
    - Reliability
    - Performance
    - Collaboration
3. When should git revert be preferred over git reset?

Task 9: Visualization

1. Display the commit graph using:

    git log --graph --oneline --all

2. Interpret the branching and merging structure.

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.

2. The report should contain each asked deliverables.

3. The PDF name should be in the form "StudentName_SAPID".